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42. State the third law of thermodynamics.

43. Write down the Born-Haber cycle for the formation of $CaCl_2$

44. Identify the state and path functions out of the following: a)Enthalpy b)Entropy c) Heat d) Temperature e) Work f)Free energy.

45. State the various statements of second law of thermodynamics.

46. What are spontaneous reactions? What are the conditions for the spontaneity of a process?

47. List the characteristics of internal energy.

48. Explain how heat absorbed at constant volume is measured using bomb calorimeter with a neat diagram.

49. Calculate the work involved in expansion and compression process.

50. Derive the relation between ΔH and ΔU for an ideal gas. Explain each term involved in the equation.

51. Suggest and explain an indirect method to calculate lattice enthalpy of sodium chloride crystal.

52. List the characteristics of Gibbs free energy.

53. Calculate the work done when 2 moles of an ideal gas expands reversibly and isothermally from a volume of 500 ml to a volume of 2 L at 25°C and normal pressure.

54. In a constant volume calorimeter, 3.5 g of a gas with molecular weight 28 was burnt in excess oxygen at 298 K. The temperature of the calorimeter was found to increase from 298 K to 298.45 K due to the combustion process. Given that the calorimeter constant is 2.5 kJ K^{-1}. Calculate the enthalpy of combustion of the gas in kJ mol^{-1}.

55. Calculate the entropy change in the system, and surroundings, and the total entropy change in the universe during a process in which 245 J of heat flow out of the system at 77°C to the surrounding at 33°C.

56. 1 mole of an ideal gas, maintained at 4.1 atm and at a certain temperature, absorbs heat 3710J and expands to 2 litres. Calculate the entropy change in expansion process.

57. 30.4 kJ is required to melt one mole of sodium chloride. The entropy change during melting is 28.4 JK^{-1} mol^{-1}. Calculate the melting point of sodium chloride.

58. Calculate the standard heat of formation of propane, if its heat of combustion is −2220.2 kJ mol^{-1}. the heats of formation of CO_2(g) and H_2O(l) are −393.5 and −285.8 kJ mol^{-1} respectively.